DIPLOMA PROGRAMME PRACTICE / SL / SET A

# Economics / Paper 2

**Question booklet · 105 minutes · 40 marks**

Choose one of the two cases and answer all parts (a) to (g) of that case. A calculator is permitted. Use the extracts and data in your answers. Show workings and units, and draw fully labelled diagrams on separate answer paper.

QUESTION 1 / 40 MARKS

## Water security and rural incomes in Morocco

Text A — Competing uses. Repeated drought has made the reliability of Morocco's water supply an economic issue for cities and farms. The national response includes better irrigation management and additional supply. In the practice district below, groundwater is accessible to several growers: withdrawing water leaves less for neighbours, while monitoring each well is expensive. A posted charge currently applies to metered withdrawals only.

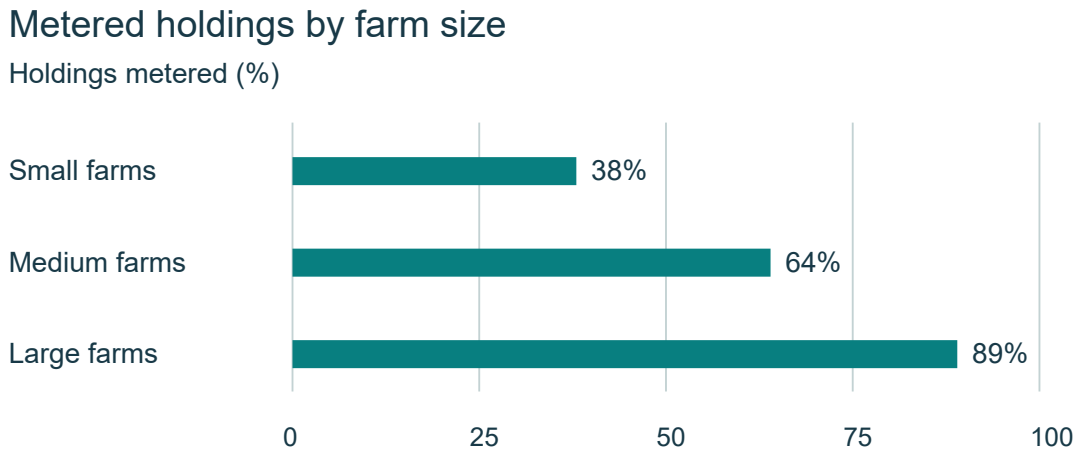
Text B — Who can adjust? Large orchards have credit and can buy drip equipment. Some tenant farmers hold one-season leases and cannot borrow against the land. Water-saving equipment can reduce withdrawals per hectare, but profitable orchards may expand planted area. A proposed charge would fund meters and a household connection grant. The alternative is a uniform subsidy on drip equipment. The treasury can fund only one new programme this year.

Text C — A pilot. The authority tested a higher charge on the same group of metered farms. Rainfall and crop prices were approximately unchanged. The neighbouring unmetered district was not in the trial. All district quantities and prices below are constructed for this exercise.

| Pilot measure | Before | After |
|---|---|---|
| Water charge (dirhams per m³) | 1.20 | 1.50 |
| Metered withdrawals (million m³ per season) | 40 | 36 |
| Hectares planted in the pilot | 10000 | 10500 |

Background reading: World Bank, Morocco water security and resilience programme (2023). All district or scenario data and policy proposals are constructed for this paper.

FIGURE 1




Constructed district data for this case.

**a(i)** Define a common pool resource in the context of the aquifer in Text A. **[2]**

**a(ii)** Define opportunity cost as it applies to the treasury's choice in Text B. **[2]**

**b(i)** Calculate the percentage change in metered withdrawals and the percentage change in the water charge, using initial values. **[3]**

**b(ii)** Calculate the revenue collected from the after-pilot charge, assuming all metered withdrawals are billed and paid. **[2]**

**c** Using a demand and supply diagram, explain how a per-unit abstraction charge paid by growers could affect the supply of orchard fruit. **[4]**

**d** Explain why credit constraints may cause a uniform equipment subsidy to benefit owners of large orchards more than tenant farmers. **[4]**

**e** Using a production possibilities diagram, explain how a lasting deterioration in reliable water supply can change the attainable combination of farm and other output. **[4]**

**f** Explain why a reduction in water used per hectare may fail to reduce total withdrawals in the long run. **[4]**

**g** Using the texts, data and your knowledge of economics, evaluate whether Morocco's practice district should prioritise a metered abstraction charge or an equipment subsidy to protect groundwater and rural livelihoods. **[15]**

QUESTION 2 / 40 MARKS

# Chile and a copper revenue shock

Text A — A volatile tax base. Copper is an important export for Chile. Its fiscal framework and stabilisation funds illustrate how spending can be separated from temporary commodity revenue. In this constructed budget exercise, a lower world copper price reduces profits and royalties. A mining town reports fewer orders for transport and repair work while its stock of machinery is largely unchanged.

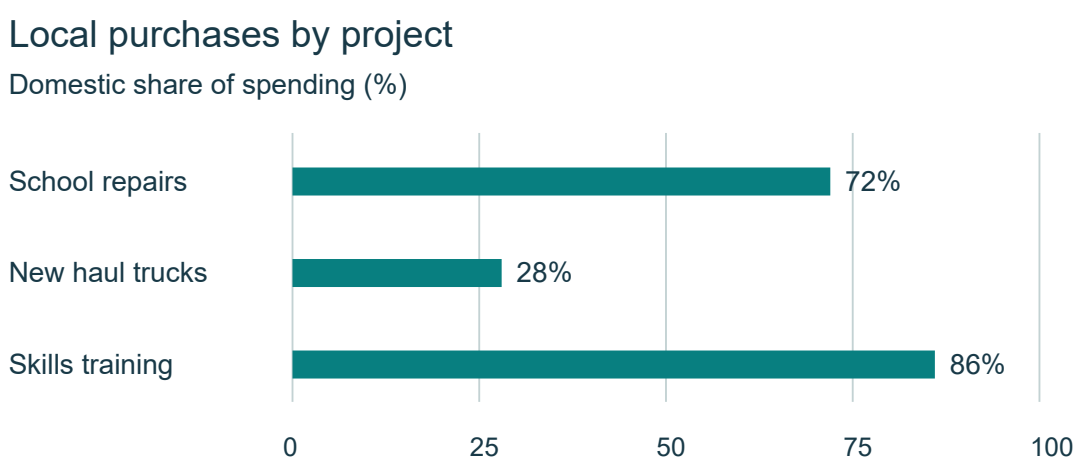
Text B — Three pressures on the budget. A finance committee can draw on savings to maintain school repairs or postpone the work for a year. The repair contracts employ local workers, although equipment is imported. A separate proposal would fund skills training for workers moving into non-mining industries. Training takes two years and vacancies are concentrated away from the town. The treasury cannot fund every commitment indefinitely.

Text C — A price-taker. The export office assumes that shipment volume below was fixed under contract for this year. Only a fraction of export receipts enters the public budget. Its operating spending forecast was prepared before the price fall. The figures are illustrative, not reported Chilean accounts.

| Annual indicator | Baseline | Shock |
|---|---|---|
| Copper price (US dollars per tonne) | 8400 | 7000 |
| Exports (million tonnes) | 4.0 | 4.0 |
| Public revenue (billion US dollars) | 80 | 75 |
| Public expenditure (billion US dollars) | 82 | 82 |
| Population (million) | 20 | 20 |

Background reading: IMF, Fiscal Considerations in Managing Stabilization Funds in Chile (2023). All district or scenario data and policy proposals are constructed for this paper.

FIGURE 1




Constructed district data for this case.

**a(i)** Define an export. **[2]**

**a(ii)** Define a budget deficit. **[2]**

**b(i)** Calculate copper export revenue in each scenario and the change, in billions of US dollars. **[3]**

**b(ii)** Calculate the shock-year budget deficit per resident. **[2]**

**c** Using an AD–AS diagram, explain how the loss of export receipts could reduce real output in the short run. **[4]**

**d** Explain why the fall in government revenue need not equal the fall in copper export revenue. **[4]**

**e** Using a labour market diagram for local repair workers, explain a possible effect of postponing school repairs. **[4]**

**f** Explain why improving occupational mobility may be insufficient to secure employment for the mining town's displaced workers. **[4]**

**g** Using the texts, data and your knowledge of economics, evaluate the use of a stabilisation fund to maintain public spending after the fall in the copper price. **[15]**